



Decentralizing Agriculture

A shift in the industrial paradigm of food production

Marco Zolla Salazar

AuthorId

Faculty

Major

bei

Employer

Kontakt:

marco.zolla@proton.me

ContactTitle

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Project vision	3
2	The problems of industrial agriculture	4
2.1	Land Use	4
2.2	Biodiversity loss	5
2.3	Soil degradation and erosion	5
2.4	Fertilizer use	6
2.5	Pesticide use	7
2.6	Monoculture	8
2.6.1	The monoculture effect: genetic vulnerability to disease and pest outbreaks	8
2.6.2	Historical and contemporary case studies	9
2.6.3	Soil degradation	9
2.6.4	Yield penalties and the diversification dividend	10
2.6.5	Diversification as a systemic alternative	11
2.6.6	Structural drivers: market concentration and monoculture	11
2.7	Freshwater depletion	11
2.8	Greenhouse gas emissions	12
2.9	Produce transportation, storage and disposal	12
2.10	The existential dependency on a few producers	12
2.11	Suppression of dietary diversity and nutritional quality	13
	References	14

Abstract

Industrial agriculture is responsible for the majority of global biodiversity loss, roughly one third of all greenhouse gas emissions, and the creation of fragile, highly concentrated food supply chains on which billions of people existentially depend. This paper proposes an open-source controlled environment agriculture (CEA) platform designed to address these problems through decentralisation, automation, and a shared data commons.

The system comprises a sensor array and camera mounted in a small-scale greenhouse, a microcontroller running a PID control loop over environmental actuators, a single-board computer performing edge ML inference, and an optional connection to a cloud-based data flywheel where growth models are continuously retrained on anonymised telemetry from the global network of deployed nodes. A computer vision pipeline provides growth metrics, disease diagnosis, and species recognition. A climate-aware species recommendation engine minimises energy consumption by matching recommended crops to the local ambient environment. All hardware designs, firmware, software, trained models, and collected data are released under open licences, enabling a virtuous cycle: the more nodes deployed, the richer the shared dataset, and the more accurate the growth optimisation.

The project does not aim to replace field agriculture wholesale. Its niche is high-value, perishable, water-intensive crops grown in proximity to consumers — the segment where the environmental and supply-chain benefits of CEA are largest and most defensible. Within that niche, we argue the platform can meaningfully reduce land use pressure, water consumption, fertiliser runoff, pesticide pollution, transport emissions, food waste, and supply-chain concentration, while enabling a diversity of crops that the industrial monoculture model structurally cannot support.

1 Introduction

1.1 Motivation

The global food system is in a state of structural crisis. Industrial agricultural production — the dominant paradigm for the past century — has delivered extraordinary increases in caloric output, but at costs that are becoming impossible to ignore: ecosystem collapse, soil exhaustion, water depletion, climate destabilisation, and the creation of food supply chains so concentrated that a single geopolitical shock can threaten the nutrition of hundreds of millions of people.

At the same time, the tools to build something better have never been more accessible. Capable micro-controllers cost a few dollars. Machine learning frameworks are open source. Camera modules capable of running computer vision inference are commodity hardware. The question is no longer whether small-scale, intelligent, automated food production is technically feasible — it is — but whether it can be made cheap enough, robust enough, and open enough to displace the incumbent system at meaningful scale.

This paper describes a platform designed to make that possible.

1.2 Project vision

The project’s core proposition is that a network of many small, locally operated, data-connected greenhouses can collectively outperform a system of few large, centralised farms — on environmental metrics, on resilience, and eventually on cost — provided that the intelligence required to operate them efficiently is shared as a common good rather than locked up in proprietary systems.

The platform is designed around four interlocking principles:

1. **Edge-first control.** All critical control logic runs locally and offline. The system operates without internet connectivity. The cloud connection is optional and asynchronous, used only for learning and coordination, never for real-time operation.
2. **Open everything.** Hardware reference designs, firmware, software, trained models, and the collected dataset are all released under maximally permissive open licences. Anyone can deploy, modify, and contribute back.
3. **Data flywheel.** Every deployed node, with the user’s consent, contributes anonymised telemetry to a shared dataset. Models trained on this dataset are continuously improved and pushed back to all nodes via over-the-air updates. The network becomes smarter as it grows.
4. **Climate-adapted intelligence.** The system does not prescribe a fixed set of crops. A climate-aware recommendation engine suggests species whose optimal growing conditions most closely match the local ambient environment, minimising the energy required to close the gap between what the environment provides and what each crop needs.

2 The problems of industrial agriculture

Although the rise of the modern industrial structure of agricultural production has given many societies the ability to forestall famine and thrive in ways long thought impossible, many of its social and ecological repercussions have revealed themselves to be potentially dangerous in the best scenario and significant contributors to an existential crisis in the worst. Every step throughout the production process results in ecological problems of varying severity, from the clearing of the land to the delivery of the final product, each impacting the surrounding ecosystems and underlying soils in different ways. It is a production model that forces the whole population into the dependence of a few producers, who have increasingly incorporated more and more monoculture practices, thus making the food supply more vulnerable to collapse in the case of an uncontrolled pest. The following sections document each of these problems in turn, drawing on recent peer-reviewed literature and institutional reports.

2.1 Land Use

[SOURCES CHECKED for factuality, need to check that the reported page is the correct one]

The prevailing paradigm of industrial agricultural production requires the use of vast swathes of land, along with a considerable amount of water, energy and other resources, which inevitably ends up homogenizing the local flora and destroying the habitat of the local fauna, leading to an overall loss of biodiversity. The agricultural expansion that the current system entails is indeed one of the main drivers of global biodiversity loss [1, p. 412], [2, p. 199], and is the most widespread form of land-use change, with cropping and animal husbandry covering over one third of the terrestrial land surface [1, p. XVI]. More than 90% of global biodiversity impacts from land-use change are attributable to agriculture, with crop cultivation accounting for 72% and pastures for 21% — dwarfing the combined impact of mining, urban development, and all other industries [3].

This geographical expansion of the agricultural sector has been a constant for decades, having grown by 78 million ha (780,000 km²) between 2001 and 2023, mostly driven by an increase in permanent crops such as cocoa, oil palm or coffee [4]. All of these are used intensively for monoculture, as are feedstock crops which by themselves account for one third of the total crops produced globally [5, p. 22]; this particular form of agriculture, which consists in the the use of a single crop for large productions, is another potent driver of biodiversity and habitat loss which will be explored in a subsequent section.

The extension of anthropogenic drivers of change has led to as little as 13% of the ocean and 23% of the land falling under the classification of ‘wilderness’. These few remaining areas that lie still outside of humanity’s sphere of influence tend to be remote or unproductive, like tundras or oceanic gyres, the most hospitable biomes having been almost or completely modified by humans in most regions [1, p. 206]. With the transformation for human convenience of the landscapes of the world that are the most accommodating to life, displaced wild populations are left without a habitat.

The impact of this intrusion into the wilderness has also far-reaching consequences as well. According to the FAO’s State of Food and Agriculture report [6, p. 4], agriculture currently drives 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area. This leads

to changes in both global and local climates through two main mechanisms: the release of stored carbon produces a robust warming signal worldwide, while the removal of forest cover changes surface albedo, evapotranspiration and canopy roughness, which produces a global warming effect for deforestation in the tropics, and a cooling effect at higher latitudes. At the local scale, forest loss almost always raises surface temperatures, with typical increases between 0.2 °C - 2 °C, depending on latitude [7].

A pan-tropical synthesis of deforestation data found that between 90 and 99% of all tropical deforestation between 2011 and 2015 was directly or indirectly driven by agricultural expansion, with 6.4–8.8 million hectares cleared per year [8]. Crucially, only 45–65% of that deforested land was subsequently converted into productive agricultural use, meaning that a substantial fraction of tropical forest destruction delivers no lasting food-production benefit.

2.2 Biodiversity loss

[SOURCES CHECKED for factuality, need to check that the reported page is the correct one]

Beyond direct habitat destruction, the intensification of agriculture has caused cascading losses across all taxa. The spread and intensification of agriculture over the past half century — through greatly expanded scales, monoculturing, increasing pesticide and fertiliser application, and the elimination of hedgerows and other wildlife habitat — is directly related to major documented declines in insect biomass and diversity [9]. At the global scale, agriculture alone has been identified as a threat to 24,000 of the 28,000 (86%) species currently at risk of extinction [10].

The impact reaches beyond field boundaries. A 27-year study using standardised traps across 63 German nature protection areas found a seasonal decline of 76% and a mid-summer peak decline of 82% in total flying insect biomass, with losses apparent regardless of habitat type and implying landscape-scale drivers rooted in agricultural intensification [11]. This decline is multi-causal and compounding: habitat loss, agricultural conversion, climate change, and pesticide exposure co-occur and interact in intensively farmed landscapes, with annual abundance declines frequently around 1–2% across taxa [12]. Across a systematic review of 73 reports, approximately 41% of insect species are declining and a third are threatened with extinction; the pace of local insect species extinction is eight times higher than that of vertebrates, with total insect biomass estimated to be declining at approximately 2.5% per year globally. [13].

2.3 Soil degradation and erosion

[SOURCES CHECKED for factuality, need to check that the reported page is the correct one]

Soils underpin the production of ecosystem goods and services that sustain both natural habitats and human societies. They host the full spectrum of biodiversity, from bacteria and fungi to plants and macro-fauna; and this biotic community drives the soil functions (nutrient cycling, water regulation, carbon sequestration, etc.) that generate those services. Those functions are controlled primarily by the soil’s chemical composition, biological activity, and physical structure. When any of these attributes deteriorate, the soil’s ability to deliver essential ecosystem services, such as food production, water purification, and climate regulation, is compromised or lost [14, pp. 4–5].

Industrial tillage-based agriculture destroys soil at a rate that outpaces natural soil formation by orders of magnitude. Erosion rates from conventionally ploughed agricultural fields average one to two orders of magnitude greater than rates of soil production, erosion under native vegetation, and long-term geological erosion — making conventional plow-based agriculture fundamentally unsustainable over civilizational timescales [15]. Industrial farming practices such as mono-cropping, use of organic fertilizers, and removal of crop residues are a primary cause of the decline in soil organic matter, which is essential to long-term soil productivity [16].

The IPCC’s Special Report on Climate Change and Land estimates that cropland soils have lost 20–60% of their organic carbon content relative to pre-cultivation states, and that soils under conventional agriculture continue to act as a net source of greenhouse gas emissions [17].

A second-order meta-analysis synthesizing 230 first-order meta-analyses encompassing more than 25,000 primary studies confirms that land conversion for crop production is a consistently net-negative driver for soil organic carbon (SOC), with mean SOC losses of approximately 25% following conversion from forest or wetland, and 16% from grassland, though most studies underestimate actual losses because they measure only over years to decades, whereas full equilibration after land-use change may take around 80 years. This matters because SOC is foundational to soil health, food production, and climate regulation: global SOC stocks to 2 m depth are estimated at approximately 2,400 Gt C — three times the carbon currently held in the atmosphere — meaning even small proportional losses translate into significant atmospheric CO₂ increases. SOC restoration is possible, but it is understood to be generally slower than depletion [18].

2.4 Fertilizer use

[SOURCES CHECKED]

Excess nitrogen and phosphorus from agricultural fertilizers leach into waterways, triggering eutrophication — a process in which accelerated algal growth reduces water clarity and, when these blooms die, microbial decomposition severely depletes dissolved oxygen, creating hypoxic or anoxic dead zones that eliminate habitat for aquatic invertebrates, fish, and other organisms, driving localized biodiversity collapse. Cultural eutrophication of this kind has been identified as a leading cause of freshwater and coastal ecosystem impairment worldwide. [19], [20]

Nearly 50% or more of applied nitrogen is lost to the environment through various mechanisms [21], [22]. Eutrophication caused by over-enrichment with phosphorus and nitrogen is a widespread problem in rivers, lakes, estuaries, and coastal waters, leading to toxic algal blooms, oxygen loss, fish kills, loss of seagrass beds, and biodiversity loss [23]. [TODO: this is a bit redundant, rephrase]

Excess nitrogen in agricultural soils undergoes microbial conversion to nitrous oxide (N₂O) — a greenhouse gas with a global warming potential 273 times that of CO₂ over 100 years [24]. Global anthropogenic N₂O emissions increased by 40% between 1980 and 2020, driven primarily by agricultural nitrogen use, and atmospheric concentrations have now exceeded all IPCC-projected scenarios, threatening Paris Agreement targets [25].

Critically, the relationship between nitrogen application and N₂O emissions is not linear but accelerating: as fertilizer inputs increase beyond crop demand, the emission factor itself grows, meaning that each

additional unit of nitrogen applied in excess generates a disproportionately larger N_2O penalty. As a consequence, small reductions in over-application can yield emissions reductions substantially larger than linear models — such as the IPCC’s default 1% emission factor — would predict.[26].

2.5 Pesticide use

[SOURCES CHECKED]

Pesticides contaminate soil, water, and air well beyond the target application zone, harming non-target organisms throughout the food web. Continuous and excessive pesticide use leads to the accumulation of toxic chemicals in soil, harming beneficial organisms such as earthworms and soil microorganisms, disrupting soil fertility and overall ecosystem health [27]. An estimated 64% of the world’s agricultural land — covering approximately 24.5 million km^2 — has been affected by pesticide pollution, and water resources across South Africa, China, India, Australia, and Argentina are heavily contaminated with pesticide residues [28].

A 2023 meta-analysis synthesizing 54 studies found that pesticides consistently reduced both the abundance and diversity of soil fauna communities. The most damaging scenarios involved combinations of multiple substances, broad-spectrum compounds, and insecticides, which significantly decreased soil invertebrate diversity even at label-recommended application rates. Critically, no evidence was found that effects diminish over time: short- and long-term studies exhibited equivalent mean effect sizes, suggesting that repeated, realistic pesticide applications may pose a persistent threat to the soil invertebrate communities that underpin ecosystem functioning. [29].

One of the most publicly visible consequences of the use of pesticides in industrial agriculture is the elevated loss rate of pollinator colonies. Neonicotinoid insecticides — a class strongly implicated in bee population declines, which is commonly applied as seed treatments [30] — are systemic compounds that translocate throughout treated plants, including into pollen, nectar, and guttation fluids, thereby exposing foraging pollinators through multiple routes of contact [31]. Multiple interacting stressors — encompassing parasites, pathogens, nutritional deficiency, and pesticide exposure — represent the most probable drivers of elevated colony loss rates [30], [31]. Furthermore, the demand for insect pollination of crops has approximately tripled over the past fifty years, raising concern that continued pollinator decline could jeopardize the stability of agricultural food production [30].

The harm does not stop at the field boundary. Systematic reviews of epidemiological and toxicological evidence consistently link chronic pesticide exposure to a wide spectrum of non-communicable diseases, including cancer, neurological disorders, and endocrine disruption [32]. A 2023 review of 63 epidemiological studies found strong human evidence for associations between pesticide exposure and various types of cancer, specifically acute myeloid leukemia and colorectal cancer, with several high-quality prospective studies observing dose-response relationships; the authors concluded that existing evidence is already sufficient to justify regulatory action [33]. Organochlorine and organophosphate compounds present in pesticides can mimic or antagonize endogenous hormones, with documented associations including breast and prostate cancers, reproductive disorders, and developmental effects in children exposed *in utero* [32], [33].

The severity of these harms is not an unavoidable cost of pest management: it is in significant part a structural consequence of the scale and indiscriminate application methods characteristic of industrial

open-field agriculture, where large-scale monoculture systems drive heightened pest pressure and perpetuate reliance on broad-spectrum chemical inputs [34], [35]. Controlled-environment production systems substantially reduce this reliance by physically excluding the pests, pathogens, and adverse weather conditions that make broad-spectrum pesticide and fertilizer application necessary in open-field contexts, thereby minimizing the need for chemical intervention at its source rather than at the margin [36], [37].

2.6 Monoculture

Monoculture — the cultivation of a single crop species, or a narrow set of genetically near-identical cultivars, over large and continuous areas — is not merely one practice among many in industrial agriculture; it is its organizing principle. The economic logic of scale, mechanization, and commodity-market integration pushes production towards ever-greater uniformity, with cascading consequences for soil health, pest and disease dynamics, ecosystem resilience, and food-system stability. The following subsections document each of these dimensions in turn.

2.6.1 The monoculture effect: genetic vulnerability to disease and pest outbreaks

A fundamental principle of epidemiology and ecology is that the severity and extent of a disease or pest outbreak depends on host population density [38] and genetic uniformity [39], [40]— and monoculture maximizes both. Genetically homogeneous host populations are more vulnerable to infection because parasites that successfully exploit one individual can readily transmit to genetically similar neighbors, encountering no variation in resistance mechanisms. This phenomenon — termed the “monoculture effect” — has been empirically validated across a wide range of host–parasite systems. A meta-analysis of 23 studies in non-crop systems found that high host genetic diversity significantly reduced parasite success, confirming that the principle extends well beyond agriculture [41]. A subsequent and larger meta-analysis synthesizing 102 studies comprising over 2,000 effect sizes validated the protective effect of genetic diversity across both crop and non-crop systems, finding that crop polycultures experienced approximately 50% less parasitism than monocultures on average [42].

The converse — that genetic diversity suppresses disease — has been demonstrated at operational scale. In a landmark field experiment in Yunnan Province, China, genetically diversified rice plantings across first five, then ten townships showed that disease-susceptible varieties planted in mixtures with resistant varieties suffered 94% less rice blast (the major disease of rice) severity than when grown in monoculture, and yielded 89% more. The results were so dramatic that fungicidal sprays were eliminated entirely by the end of the two-year programme [39]. This experiment remains one of the most compelling demonstrations that intraspecific crop diversification can function as a practical, large-scale disease management strategy.

In monoculture systems, the genetic uniformity of the crop creates conditions in which pest and pathogen populations can spread rapidly and exploit the entire stand, since overcoming the defences of one plant effectively compromises all plants [43]. Several pests reach higher population densities in monocultures precisely because the consistent and abundant host supply supports unimpeded reproduction [43], [44], while the reduction in habitat diversity suppresses the natural enemy communities that would otherwise regulate outbreaks [44]. The resulting dependence on broad-spectrum pesticides to compensate for the loss

of ecological pest control creates a self-reinforcing cycle: pesticide use further degrades the biodiversity that would limit pest pressure, necessitating still more chemical intervention.

2.6.2 Historical and contemporary case studies

The vulnerability inherent in genetic uniformity is not theoretical. It has repeatedly materialized in catastrophic losses throughout the history of modern agriculture.

The 1970 Southern Corn Leaf Blight—caused by a single cytoplasmic male-sterility trait (T-cytoplasm) shared by over 85% of US hybrid maize varieties [45]—destroyed 15% of the American crop in a single season and caused approximately \$1 billion in losses [45], [46], a direct consequence of genetic homogeneity [46].

Global banana production depends overwhelmingly on a single cultivar — the Cavendish subgroup — which represents over 40% of global production and makes up almost all of the export trade, with an additional 20% of production consisting of closely related varieties sharing a very similar genetic background. The Cavendish itself rose to dominance only because its predecessor, the Gros Michel, was decimated by *Fusarium* wilt Race 1 during the first half of the twentieth century — a disaster enabled by the genetic uniformity of the Gros Michel monoculture [47]. History is now repeating: a new strain, *Fusarium oxysporum* f.sp. *cubense* Tropical Race 4 (TR4), is spreading through Cavendish plantations worldwide: since first being identified in Taiwan in the 1960s, TR4 has expanded to at least 23 countries as of 2024, including Colombia and Peru, the heart of the Latin American export industry [48]. The pathogen can survive in soil for up to 30 years in the form of chlamydospores, spreads through contaminated soil, water, and planting material, and there are currently no market-acceptable resistant cultivar replacements [49]. The global banana industry thus faces an existential threat that is a direct structural consequence of monoculture production: an entire sector built on a single clone, with no viable fallback.

These are not isolated events. The Irish Potato Famine of the 1840s, the coffee rust epidemic that devastated Sri Lanka’s industry in the 1860s–1880s, and recurring wheat rust pandemics all illustrate how genetic uniformity at scale can leave entire production systems vulnerable to a single pathogen [50]–[52].

2.6.3 Soil degradation

Continuous monoculture degrades the very substrate upon which it depends. Growing the same crop repeatedly depletes specific nutrients, disrupts soil microbial communities and their functional capacity, and diminishes soil organic matter (SOM) and active carbon, which are essential to long-term soil productivity [53].

The impact on soil biota is also substantial. Monoculture systems exhibit reduced soil microbial diversity and altered community composition compared to diverse cropping systems [54]. The narrowing of root exudate chemistry under continuous cropping selects against a broad range of beneficial rhizosphere taxa — including disease-suppressive bacteria such as *Bacillus*, *Paenibacillus*, and *Lysinibacillus* — while favouring soil-borne pathogens such as *Fusarium* [55], [56]. Conversely, crop rotation increases microbial biomass carbon and nitrogen by around 13–16% and raises bacterial diversity relative to continuous monoculture [54], [57]. These biological improvements translate into measurable structural gains: a meta-analysis of 2,199

paired observations found that rotation improved macroaggregate stability¹ by 7–9% relative to continuous monoculture, with initial soil organic carbon identified as the predominant soil-level driver of this effect [59] — consistent with the established role of microbial biomass carbon and nitrogen in aggregate formation [60], and with the inverse losses that monoculture’s suppression of carbon inputs and microbial communities imposes.

The resulting erosion is not a marginal effect. As documented previously (Section 2.3), erosion rates from conventionally ploughed monoculture fields average one to two orders of magnitude greater than rates of soil production — making the practice fundamentally unsustainable over civilisational timescales [15]. Where monocultures displace the diverse root architectures and ground cover of polycultures, rain runoff increases [61] and the soil’s capacity to retain water diminishes [62], driving a secondary dependence on irrigation infrastructure [63].

[SOURCES CHECKED UP TO HERE]

2.6.4 Yield penalties and the diversification dividend

Far from maximizing output, monoculture imposes a persistent yield penalty relative to diversified systems. In the US, well-studied benefits of crop rotation often lead to 5–10% higher maize yields in even just a two-crop rotation of maize and soybean, despite the monoculture system typically requiring more inputs [64]. A global meta-analysis synthesizing 3,663 paired field-trial yield observations (1980–2024) found that crop rotation increased subsequent crop yield by 16–23% on average (depending on whether the pre-crop was a non-legume or legume), and that considering the entire rotation sequence, total yields, dietary energy, protein, and revenue all increased by 14–27% relative to continuous monoculture [65]. Notably, win-win relationships among yield, nutrition, and revenue were consistently more frequent (33–54%) than trade-offs.

The benefits extend beyond yield quantity to yield stability — a dimension of critical importance for food security. An analysis of five decades of FAO data across 91 nations found that greater effective crop diversity at the national level was associated with significantly increased temporal stability of total national harvest. Countries with the lowest crop diversities experienced a severe food shortage (defined as a $\geq 25\%$ decline in total yield) approximately every eight years; those with the highest diversities experienced one roughly every hundred years [66]. The stabilizing effect of diversity was comparable in magnitude to the destabilizing effect of precipitation variability, indicating that crop diversification provides a quantitatively meaningful buffer against climatic shocks.

Long-term field experiments across North America further confirm that more diverse rotations increased maize yields in 7 of 11 sites during the most stressful growing conditions, with the largest resilience benefits observed where diversity differences were greatest [64]. This evidence is consistent with the broader ecological principle — well established in grassland biodiversity experiments — that diversity increases the stability and productivity of biological communities.

¹Macroaggregates are soil clumps >0.25 mm held together by organic matter and microbial binding agents. Their stability — resistance to breakdown by water and physical disturbance — governs erosion susceptibility, water infiltration, and the physical protection of organic carbon from decomposition [58], [59].

2.6.5 Diversification as a systemic alternative

The empirical case against monoculture is not limited to individual metrics. A second-order meta-analysis reviewing 98 first-order meta-analyses and encompassing over 5,160 original studies (41,946 comparisons between diversified and simplified practices) found that agricultural diversification enhances biodiversity, pollination, pest control, nutrient cycling, soil fertility, and water regulation without compromising crop yields. In 63% of cases, diversification resulted in win-win outcomes for both ecosystem services and crop production simultaneously [67]. These results suggest that the simplification inherent in monoculture is not merely an environmental cost borne for the sake of production; it is a production strategy that is empirically dominated by more diverse alternatives across multiple dimensions.

Monoculture systems in Kenya and Bolivia were found to be highly susceptible to climate shocks, price volatility, soil fertility loss, and pest outbreaks — demonstrating the structural fragility of agroindustrial monocropping in developing-country contexts where resilience failures translate directly into food crises [68].

2.6.6 Structural drivers: market concentration and monoculture

The persistence of monoculture despite its well-documented costs is not primarily a matter of agronomic ignorance; it is the structural outcome of concentrated market power and the economic incentives it creates. Two companies (Syngenta Group and Bayer) control 40% of the global commercial seed market, while four firms control 62% of the world agrochemical market — market concentrations that meet the standard economic definition of oligopoly [69]. Even where many farms exist, their effective genetic choices are constrained by a handful of input suppliers, making homogenisation not merely an agronomic preference but the structural outcome of concentrated market power. The economic logic is self-reinforcing: seed companies optimize for a small number of high-yielding varieties suited to monoculture and broad-spectrum chemical regimes; farmers adopt these packages because they minimize managerial complexity; the resulting uniformity creates pest and disease pressures that can only be managed with the same companies' agrochemical products.

Breaking this cycle requires not only technical alternatives — such as the CEA platform described in this paper — but open-source alternatives that are not subject to the same market concentration dynamics.

2.7 Freshwater depletion

Agriculture is responsible for nearly 70% of global freshwater withdrawal volume, making it the most water-intensive sector by far [70]. As much as 60% of agricultural water is lost due to inappropriate irrigation systems and poor application techniques, contributing to the depletion of rivers, lakes, and underground aquifers.

The full scale of agricultural water consumption is partially obscured because it includes the depletion of non-renewable aquifers. Global groundwater depletion increased 22% in just one decade, from 240 km³/year in 2000 to 292 km³/year in 2010, with five commodity crops — wheat (22%), rice (17%), sugar (7%), cotton (7%), and maize (5%) — accounting for 83% of all groundwater depletion embedded in international food trade [71]. Unlike surface water shortages, depletion of fossil aquifers is functionally irreversible on

human timescales: some of the groundwater currently consumed to irrigate these crops fell as precipitation thousands of years ago and will not recharge within any planning horizon relevant to food security policy.

2.8 Greenhouse gas emissions

Global agrifood systems emit approximately 16.2 billion tonnes of CO₂-equivalent per year, accounting for roughly one third of total anthropogenic greenhouse gas emissions [72]. These are generated across farm production, land-use change, and the entire supply chain including processing, retail, and food disposal.

The aggregate figure masks the severity of individual gas contributions. Nitrous oxide from agricultural nitrogen use alone grew 30% between 1980 and 2020, increasing faster than any IPCC scenario had projected [73]. An independent lifecycle analysis places the total food-systems contribution at 34% of all anthropogenic greenhouse gas emissions — 18 Gt CO₂-eq/year as of 2015 — with post-farm supply-chain activities (processing, packaging, transport, retail, and disposal) already on track in many countries to exceed farm-gate and land-use-change emissions as the dominant source [74].

2.9 Produce transportation, storage and disposal

Producing, transporting, and allowing food to rot contributes more than 8–10% of global greenhouse gas emissions — nearly five times the total emissions from the entire aviation sector [75]. Food waste comprises about 24% of municipal solid waste in landfills, and due to its fast decay rate, an estimated 58% of fugitive methane emissions from municipal solid waste landfills are attributable to landfilled food waste [76]. As production efficiency increases, the relative weight of these post-farm emissions grows, making local, short-supply-chain production an increasingly important lever for climate mitigation [74].

2.10 The existential dependency on a few producers

A significant 60% of the world’s food production occurs in just five countries: China, the United States, India, Brazil, and Argentina [77]. This geographic concentration makes a substantial segment of the world’s food supply susceptible to geopolitical disruption, extreme weather, and infrastructure failure. The world has experienced three global food crises in the past 50 years, each exposing the same underlying structural feature: the global industrial food system’s reliance on a small number of staple grains produced using highly industrialised farming methods, making it susceptible to events affecting just a handful of crops or regions [78].

The concentration risk extends upstream beyond geography. In the commercial seed sector, two companies (Syngenta Group and Bayer) control 40% of the global market; in agrochemicals, four firms control 62%; similar oligopolistic concentration characterises agricultural machinery and animal pharmaceuticals [69]. This upstream concentration creates systemic risk through interlocking ownership structures that reduce effective competition and introduce single points of failure invisible in geographic analyses of production alone [79]. The existential dependency is therefore not merely on a few producing regions, but on a small number of firms that supply the inputs upon which those regions — and the rest of the world’s food supply — depend.

2.11 Suppression of dietary diversity and nutritional quality

Industrialised agriculture prioritises monoculture production, contributing to the global proliferation of ultra-processed foods and the concurrent rise in obesity and non-communicable diseases including diabetes and cardiovascular disease, alongside the drastic acceleration of ecosystem decline [80].

The scale of this homogenisation is empirically documented. An analysis of national food supply data from 152 countries over 50 years found that global diets converged significantly towards a common pattern dominated by wheat, rice, maize, and sugar, with historically diverse regions experiencing the most rapid homogenisation [81]. Modelling by Springmann et al. shows that current dietary trends — dominated by industrial monoculture commodity crops and animal products — will push multiple planetary boundaries beyond safe limits by 2050, including those for climate change, land use, freshwater use, and nitrogen pollution, even under optimistic assumptions about production efficiency improvements; diversification of diets towards plant-rich, varied foods is identified as the single highest-impact lever available to policymakers [82].

References

- [1] Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, “Global assessment report of the intergovernmental science-policy platform on biodiversity and ecosystem services,” IPBES Secretariat, Bonn, Germany, Tech. Rep., 2019, IPBES Analytical Brief, No. 107, p. 1144. DOI: 10.5281/zenodo.3831673.
- [2] European Commission: Directorate-General for Financial Stability, Financial Services and Capital Markets Union, IASA & Trinomics, *Study for a methodological framework and assessment of potential financial risks associated with biodiversity loss and ecosystem degradation – Final report*. Publications Office of the European Union, 2024. DOI: 10.2874/30768. [Online]. Available: <https://data.europa.eu/doi/10.2874/30768>.
- [3] L. Cabernard *et al.*, “Biodiversity impacts of recent land-use change driven by increases in agri-food imports,” *Nature Sustainability*, 2024. DOI: 10.1038/s41893-024-01433-4.
- [4] Food and Agriculture Organization of the United Nations, “Land statistics 2001–2023 – global, regional and country trends,” FAOSTAT Analytical Briefs, Rome, Tech. Rep. No. 107, 2025, FAO. DOI: 10.4060/cd5765en. [Online]. Available: <https://openknowledge.fao.org/handle/20.500.14283/cd5765en>.
- [5] J. Battersby, M. Belay, J. Clapp, *et al.*, “Food from somewhere: Building food security and resilience through territorial markets,” International Panel of Experts on Sustainable Food Systems (IPES-Food), Berlin, Germany, Report, Jul. 2024. [Online]. Available: <https://ipes-food.org/publications/food-from-somewhere>.
- [6] Food and Agriculture Organization of the United Nations, “The state of food and agriculture 2025 – addressing land degradation across landholding scales,” FAO, Rome, Tech. Rep., 2025, FAO publication. DOI: 10.4060/cd7067en. [Online]. Available: <https://doi.org/10.4060/cd7067en>.
- [7] D. Lawrence, M. Coe, W. Walker, L. Verchot, and K. Vandecar, “The unseen effects of deforestation: Biophysical effects on climate,” *Front. For. Glob. Change*, vol. 5, Mar. 2022. DOI: 10.3389/ffgc.2022.756115. [Online]. Available: <https://www.frontiersin.org/journals/forests-and-global-change/articles/10.3389/ffgc.2022.756115/full>.
- [8] F. Pendrill, T. A. Gardner, P. Meyfroidt, U. M. Persson, *et al.*, “Disentangling the numbers behind agriculture-driven tropical deforestation,” *Science*, vol. 377, no. 6611, 2022. DOI: 10.1126/science.abm9267.
- [9] M. L. Forister *et al.*, “Agricultural intensification and climate change are rapidly decreasing insect biodiversity,” *Proceedings of the National Academy of Sciences*, vol. 118, no. 2, 2021. DOI: 10.1073/pnas.2002548117.
- [10] Chatham House, “Food system impacts on biodiversity loss: Three levers for food system transformation in support of nature,” Royal Institute of International Affairs, 2021. [Online]. Available: <https://www.chathamhouse.org/2021/02/food-system-impacts-biodiversity-loss>.

- [11] C. A. Hallmann, M. Sorg, E. Jongejans, *et al.*, “More than 75 percent decline over 27 years in total flying insect biomass in protected areas,” *PLOS ONE*, vol. 12, no. 10, e0185809, 2017. DOI: 10.1371/journal.pone.0185809.
- [12] D. L. Wagner, E. M. Grames, M. L. Forister, M. R. Berenbaum, and D. Stopak, “Insect decline in the Anthropocene: Death by a thousand cuts,” *Proceedings of the National Academy of Sciences*, vol. 118, no. 2, e2023989118, 2021. DOI: 10.1073/pnas.2023989118.
- [13] F. Sánchez-Bayo and K. A. Wyckhuys, “Worldwide decline of the entomofauna: A review of its drivers,” *Biological Conservation*, vol. 232, pp. 8–27, 2019. DOI: 10.1016/j.biocon.2019.01.020.
- [14] Food and Agriculture Organization of the United Nations, *Revised World Soil Charter*, en. Rome, Italy: FAO, 2015.
- [15] D. R. Montgomery, “Soil erosion and agricultural sustainability,” *Proceedings of the National Academy of Sciences*, vol. 104, no. 33, pp. 13 268–13 272, 2007. DOI: 10.1073/pnas.0611508104.
- [16] H. L. Tuomisto *et al.*, “Effects of environmental change on population nutrition and health: A comprehensive framework with a focus on fruits and vegetables,” *Wellcome Open Research*, 2017. DOI: 10.12688/wellcomeopenres.11190.2. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5814744/>.
- [17] IPCC, “Special report on climate change and land,” Intergovernmental Panel on Climate Change, Tech. Rep., 2019. [Online]. Available: <https://www.ipcc.ch/srccl/>.
- [18] D. Beillouin *et al.*, “A global meta-analysis of soil organic carbon in the Anthropocene,” *Nature Communications*, vol. 14, p. 3700, 2023. DOI: 10.1038/s41467-023-39338-z.
- [19] M. F. Chislock, E. Doster, R. A. Zitomer, and A. E. Wilson, “Eutrophication: Causes, Consequences, and Controls in Aquatic Ecosystems,” *Nature Education Knowledge*, vol. 4, no. 4, p. 10, 2013. [Online]. Available: <https://www.nature.com/scitable/knowledge/library/eutrophication-causes-consequences-and-controls-in-aquatic-102364466>.
- [20] R. J. Diaz and R. Rosenberg, “Spreading dead zones and consequences for marine ecosystems,” *Science*, vol. 321, no. 5891, pp. 926–929, 2008. DOI: 10.1126/science.1156401.
- [21] J. Martínez-Dalmau, J. Berbel, and R. Ordóñez-Fernández, “Nitrogen fertilization. a review of the risks associated with the inefficiency of its use and policy responses,” *Sustainability*, vol. 13, no. 10, 2021, ISSN: 2071-1050. DOI: 10.3390/su13105625. [Online]. Available: <https://www.mdpi.com/2071-1050/13/10/5625>.
- [22] D. Coskun, D. T. Britto, W. Shi, and H. J. Kronzucker, “Nitrogen transformations in modern agriculture and the role of biological nitrification inhibition,” *Nature Plants*, vol. 3, p. 17 074, 2017. DOI: 10.1038/nplants.2017.74.
- [23] S. Carpenter *et al.*, “Nonpoint pollution of surface waters with phosphorus and nitrogen,” Ecological Society of America, 1998. [Online]. Available: <https://esa.org/wp-content/uploads/2013/03/issue3.pdf>.

- [24] P. Forster, T. Storelvmo, K. Armour, *et al.*, “The earth’s energy budget, climate feedbacks, and climate sensitivity,” in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte, P. Zhai, A. Pirani, *et al.*, Eds., Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2021, ch. 7, pp. 923–1054. DOI: 10.1017/9781009157896.009.
- [25] H. Tian, N. Pan, R. L. Thompson, *et al.*, “Global nitrous oxide budget (1980–2020),” *Earth System Science Data*, vol. 16, no. 6, pp. 2543–2604, 2024. DOI: 10.5194/essd-16-2543-2024.
- [26] I. Shcherbak, N. Millar, and G. P. Robertson, “Global metaanalysis of the nonlinear response of soil nitrous oxide (n₂o) emissions to fertilizer nitrogen,” *Proceedings of the National Academy of Sciences*, vol. 111, no. 25, pp. 9199–9204, 2014. DOI: 10.1073/pnas.1322434111. eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.1322434111>. [Online]. Available: <https://www.pnas.org/doi/abs/10.1073/pnas.1322434111>.
- [27] A. Alengebawy *et al.*, “Pesticides impacts on human health and the environment with their mechanisms of action and possible countermeasures,” *Heliyon*, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11016626/>.
- [28] F. H. M. Tang, M. Lenzen, A. McBratney, and F. Maggi, “Risk of pesticide pollution at the global scale,” *Nature Geoscience*, vol. 14, pp. 206–210, Apr. 2021. DOI: 10.1038/s41561-021-00712-5.
- [29] L. Beaumelle, L. Tison, N. Eisenhauer, *et al.*, “Pesticide effects on soil fauna communities—A meta-analysis,” *Journal of Applied Ecology*, vol. 60, no. 7, pp. 1239–1253, 2023. DOI: 10.1111/1365-2664.14437.
- [30] D. Goulson, E. Nicholls, C. Botías, and E. L. Rotheray, “Bee declines driven by combined stress from parasites, pesticides, and lack of flowers,” *Science*, vol. 347, no. 6229, p. 1255957, 2015. DOI: 10.1126/science.1255957.
- [31] A. Fairbrother, J. Purdy, T. Anderson, and R. Fell, “Risks of neonicotinoid insecticides to honeybees,” *Environmental Toxicology and Chemistry*, vol. 33, no. 4, pp. 719–731, 2014. DOI: 10.1002/etc.2527.
- [32] C. Shekhar, R. Khosya, K. Thakur, *et al.*, “A systematic review of pesticide exposure, associated risks, and long-term human health impacts,” *Toxicology Reports*, vol. 13, p. 101840, 2024, ISSN: 2214-7500. DOI: <https://doi.org/10.1016/j.toxrep.2024.101840>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2214750024002233>.
- [33] H. Cavalier, L. Trasande, and M. Porta, “Exposures to pesticides and risk of cancer: Evaluation of recent epidemiological evidence in humans and paths forward,” *International Journal of Cancer*, vol. 152, no. 5, pp. 879–912, 2023. DOI: 10.1002/ijc.34300.
- [34] M. Tudi, H. D. Ruan, L. Wang, *et al.*, “Agriculture development, pesticide application and its impact on the environment,” *International Journal of Environmental Research and Public Health*, vol. 18, no. 3, p. 1112, 2021. DOI: 10.3390/ijerph18031112.

- [35] S. R. Belmain, Y. Tembo, A. G. Mkindi, S. E. J. Arnold, and P. C. Stevenson, “Elements of agroecological pest and disease management,” *Elementa: Science of the Anthropocene*, vol. 10, no. 1, p. 00099, 2022. DOI: 10.1525/elementa.2021.00099.
- [36] A. P. Boyd, P. Zankowski, R. Wheeler, *et al.*, “Controlled environment agriculture: An opportunity to strengthen interagency research collaboration in the US government,” *PNAS Nexus*, vol. 4, no. 6, pgaf155, 2025. DOI: 10.1093/pnasnexus/pgaf155.
- [37] S. Oh and C. Lu, “Vertical farming - smart urban agriculture for enhancing resilience and sustainability in food security,” *The Journal of Horticultural Science and Biotechnology*, vol. 98, no. 2, pp. 133–140, 2023. DOI: 10.1080/14620316.2022.2141666.
- [38] R. M. Anderson and R. M. May, “The invasion, persistence and spread of infectious diseases within animal and plant communities,” *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, vol. 314, no. 1167, pp. 533–570, 1986. DOI: 10.1098/rstb.1986.0072.
- [39] Y. Zhu, H. Chen, J. Fan, *et al.*, “Genetic diversity and disease control in rice,” *Nature*, vol. 406, pp. 718–722, 2000. DOI: 10.1038/35021046.
- [40] C. C. Mundt, “Use of multiline cultivars and cultivar mixtures for disease management,” *Annual Review of Phytopathology*, vol. 40, pp. 381–410, 2002. DOI: 10.1146/annurev.phyto.40.011402.113723.
- [41] A. K. E. Ekroth, C. Rafaluk-Mohr, and K. C. King, “Host genetic diversity limits parasite success beyond agricultural systems: A meta-analysis,” *Proceedings of the Royal Society B: Biological Sciences*, vol. 286, no. 1911, p. 20191811, 2019. DOI: 10.1098/rspb.2019.1811.
- [42] A. K. Gibson and A. E. Nguyen, “Does genetic diversity protect host populations from parasites? A meta-analysis across natural and agricultural systems,” *Evolution Letters*, vol. 5, no. 1, pp. 16–32, 2021. DOI: 10.1002/evl3.206.
- [43] J. F. Tooker and S. D. Frank, “Genotypically diverse cultivar mixtures for insect pest management and increased crop yields,” *Journal of Applied Ecology*, vol. 49, no. 5, pp. 974–985, 2012. DOI: 10.1111/j.1365-2664.2012.02173.x.
- [44] D. A. Andow, “Vegetational diversity and arthropod population response,” *Annual Review of Entomology*, vol. 36, no. 1, pp. 561–586, 1991. DOI: 10.1146/annurev.en.36.010191.003021.
- [45] H. A. Bruns, “Southern corn leaf blight: A story worth retelling,” *Agronomy Journal*, vol. 109, no. 4, pp. 1218–1224, 2017. DOI: 10.2134/agronj2017.01.0006.
- [46] L. A. Tatum, “The southern corn leaf blight epidemic,” *Science*, vol. 171, no. 3976, pp. 1113–1116, 1971. DOI: 10.1126/science.171.3976.1113.
- [47] A. Drenth and G. Kema, “The vulnerability of bananas to globally emerging disease threats,” *Phytopathology*, vol. 111, no. 12, pp. 2146–2161, 2021, PMID: 34231377. DOI: 10.1094/PHYTO-07-20-0311-RVW.
- [48] T. Munhoz, J. Vargas, L. Teixeira, C. Staver, and M. Dita, “Fusarium Tropical Race 4 in Latin America and the Caribbean: Status and global research advances towards disease management,” *Frontiers in Plant Science*, vol. 15, p. 1397617, 2024. DOI: 10.3389/fpls.2024.1397617.

- [49] J. M. Roberts, L. C. Carvalhais, C. O'Dwyer, V. A. Rincón-Flórez, and A. Drenth, "Diagnostics of Fusarium wilt in banana: Current status and challenges," *Plant Pathology*, vol. 73, no. 4, pp. 760–776, 2024. DOI: 10.1111/ppa.13863.
- [50] K. Yoshida, V. J. Schuenemann, L. M. Cano, *et al.*, "The rise and fall of the *Phytophthora infestans* lineage that triggered the Irish potato famine," *eLife*, vol. 2, e00731, 2013. DOI: 10.7554/eLife.00731.
- [51] P. Talhinhos, D. Batista, I. Diniz, *et al.*, "The coffee leaf rust pathogen *Hemileia vastatrix*: One and a half centuries around the tropics," *Molecular Plant Pathology*, vol. 18, no. 8, pp. 1039–1051, 2017. DOI: 10.1111/mpp.12512.
- [52] R. P. Singh, P. K. Singh, J. Rutkoski, *et al.*, "Disease impact on wheat yield potential and prospects of genetic control," *Annual Review of Phytopathology*, vol. 54, pp. 303–322, 2016. DOI: 10.1146/annurev-phyto-080615-095835.
- [53] Z. H. Pervaiz, J. Iqbal, Q. Zhang, D. Chen, H. Wei, and M. Saleem, "Continuous cropping alters multiple biotic and abiotic indicators of soil health," *Soil Systems*, vol. 4, no. 4, p. 59, 2020. DOI: 10.3390/soilsystems4040059. [Online]. Available: <https://doi.org/10.3390/soilsystems4040059>.
- [54] C. Li, L. Shi, K. Wang, *et al.*, "Crop rotation differentially increases soil bacterial and fungal diversities in global croplands: A meta-analysis," *Nature Communications*, vol. 16, p. 11 686, 2025. DOI: 10.1038/s41467-025-66823-4.
- [55] P. Li, J. Liu, M. Saleem, *et al.*, "Reduced chemodiversity suppresses rhizosphere microbiome functioning in the mono-cropped agroecosystems," *Microbiome*, vol. 10, p. 108, 2022. DOI: 10.1186/s40168-022-01287-y.
- [56] Y. Zhou, Z. Yang, J. Liu, *et al.*, "Crop rotation and native microbiome inoculation restore soil capacity to suppress a root disease," *Nature Communications*, vol. 14, no. 1, p. 8126, 2023. DOI: 10.1038/s41467-023-43926-4.
- [57] Q. Liu, Y. Zhao, T. Li, L. Chen, Y. Chen, and P. Sui, "Changes in soil microbial biomass, diversity, and activity with crop rotation in cropping systems: A global synthesis," *Applied Soil Ecology*, vol. 186, p. 104815, 2023. DOI: 10.1016/j.apsoil.2023.104815.
- [58] J. M. Tisdall and J. M. Oades, "Organic matter and water-stable aggregates in soils," *Journal of Soil Science*, vol. 33, no. 2, pp. 141–163, 1982. DOI: 10.1111/j.1365-2389.1982.tb01755.x.
- [59] F. Zheng, X. Liu, W. Ding, X. Song, S. Li, and X. Wu, "Positive effects of crop rotation on soil aggregation and associated organic carbon are mainly controlled by climate and initial soil carbon content: A meta-analysis," *Agriculture, Ecosystems & Environment*, vol. 355, p. 108 600, 2023. DOI: 10.1016/j.agee.2023.108600.
- [60] M. D. McDaniel, L. K. Tiemann, and A. S. Grandy, "Does agricultural crop diversity enhance soil microbial biomass and organic matter dynamics? A meta-analysis," *Ecological Applications*, vol. 24, no. 3, pp. 560–570, 2014. DOI: 10.1890/13-0616.1.

- [61] A. D. Basche and M. S. DeLonge, “Comparing infiltration rates in soils managed with conventional and alternative farming methods: A meta-analysis,” *PLOS ONE*, vol. 14, no. 9, e0215702, 2019. DOI: 10.1371/journal.pone.0215702.
- [62] A. D. Basche and M. DeLonge, “The impact of continuous living cover on soil hydrologic properties: A meta-analysis,” *Soil Science Society of America Journal*, vol. 81, no. 5, pp. 1179–1190, 2017. DOI: 10.2136/sssaj2017.03.0077.
- [63] L. Alletto, A. Vandewalle, and P. Debaeke, “Crop diversification improves cropping system sustainability: An 8-year on-farm experiment in south-western france,” *Agricultural Systems*, vol. 200, p. 103 433, 2022. DOI: 10.1016/j.agsy.2022.103433.
- [64] T. M. Bowles, M. Mooshammer, Y. Socolar, *et al.*, “Long-term evidence shows that crop-rotation diversification increases agricultural resilience to adverse growing conditions in North America,” *One Earth*, vol. 2, no. 3, pp. 284–293, 2020. DOI: 10.1016/j.oneear.2020.02.007.
- [65] D. Beillouin *et al.*, “Crop rotations synergize yield, nutrition, and revenue: A meta-analysis,” *Nature Communications*, 2025. DOI: 10.1038/s41467-025-64567-9.
- [66] D. Renard and D. Tilman, “National food production stabilized by crop diversity,” *Nature*, vol. 571, no. 7764, pp. 257–260, 2019. DOI: 10.1038/s41586-019-1316-y.
- [67] G. Tamburini, R. Bommarco, T. C. Wanger, *et al.*, “Agricultural diversification promotes multiple ecosystem services without compromising yield,” *Science Advances*, vol. 6, no. 45, eaba1715, 2020. DOI: 10.1126/sciadv.aba1715.
- [68] S. Mukhovi and J. Jacobi, “Can monocultures be resilient? assessment of buffer capacity in two agroindustrial cropping systems in africa and south america,” *Agriculture & Food Security*, vol. 11, p. 19, 2022. DOI: 10.1186/s40066-022-00356-7.
- [69] ETC Group and GRAIN, “Top 10 agribusiness giants: Corporate concentration in food and farming in 2025,” ETC Group, Tech. Rep., 2025. [Online]. Available: <https://www.etcgroup.org/content/top-10-agribusiness-giants>.
- [70] C. Ingrao *et al.*, “Water scarcity in agriculture: An overview of causes, impacts and approaches for reducing the risks,” *Heliyon*, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10392093/>.
- [71] C. Dalin, Y. Wada, T. Kastner, and M. J. Puma, “Groundwater depletion embedded in international food trade,” *Nature*, vol. 543, pp. 700–704, 2017. DOI: 10.1038/nature21403.
- [72] Food and Agriculture Organization of the United Nations, “Greenhouse gas emissions from agrifood systems. global, regional and country trends, 2000–2022,” FAO, 2024. [Online]. Available: <https://www.fao.org/statistics/highlights-archive/highlights-detail/greenhouse-gas-emissions-from-agrifood-systems>.
- [73] H. Tian *et al.*, “A comprehensive quantification of global nitrous oxide sources and sinks,” *Nature*, vol. 586, pp. 248–256, 2020. DOI: 10.1038/s41586-020-2780-0.

- [74] M. Crippa, E. Solazzo, D. Guizzardi, *et al.*, “Food systems are responsible for a third of global anthropogenic GHG emissions,” *Nature Food*, vol. 2, pp. 198–209, 2021. DOI: 10.1038/s43016-021-00225-9.
- [75] United Nations, “Food and climate change: Healthy diets for a healthier planet,” United Nations, 2023. [Online]. Available: <https://www.un.org/en/climatechange/science/climate-issues/food>.
- [76] United States Environmental Protection Agency, “Quantifying methane emissions from landfilled food waste,” US EPA, 2023. [Online]. Available: <https://www.epa.gov/land-research/quantifying-methane-emissions-landfilled-food-waste>.
- [77] M. Masoura, “The fragile link: Supply chain disruptions and global food security,” *Food Science and Technology*, 2024. DOI: 10.1002/fsat.3803_9.x.
- [78] J. Clapp, “Concentration and crises: Exploring the deep roots of vulnerability in the global industrial food system,” *Journal of Peasant Studies*, 2022. DOI: 10.1080/03066150.2022.2129013.
- [79] J. Clapp, “The problem with growing corporate concentration and power in the global food system,” *Nature Food*, vol. 2, pp. 404–408, 2021. DOI: 10.1038/s43016-021-00297-7.
- [80] C. Rosier *et al.*, “From soil to health: Advancing regenerative agriculture for improved food quality and nutrition security,” *Frontiers in Nutrition*, 2025. DOI: 10.3389/fnut.2025.1638507. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12576041/>.
- [81] C. K. Khoury, A. D. Bjorkman, H. Dempewolf, *et al.*, “Increasing homogeneity in global food supplies and the implications for food security,” *Proceedings of the National Academy of Sciences*, vol. 111, no. 11, pp. 4001–4006, 2014. DOI: 10.1073/pnas.1313490111.
- [82] M. Springmann *et al.*, “Options for keeping the food system within environmental limits,” *Nature*, vol. 562, pp. 519–525, 2018. DOI: 10.1038/s41586-018-0594-0.